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Kerguelen: Continental Fragment or Oceanic Island?

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Kerguelen: Continental Fragment or Oceanic Island?

ABSTRACT

In order to contribute to resolution of the possibility that the Kerguelen-Gaussberg Ridge (in the south-central Indian Ocean) is a continental fragment remaining after separation of Gondwanaland, a series of analyses have been made on collections of igneous rock from the island of Kerguelen. Petrographic analyses and measurement of the major- and trace-element concentrations, Sr-isotopic ratios, K-Ar ages, and paleomagnetic properties of intrusive and extrusive bodies have been added to the known volcanic stratigraphy and some previously unpublished geochemical data, to arrive at a preferred origin for the island. It is concluded that Kerguelen is not older than late Oligocene to early Miocene in outcrop, and that it migrated southwestward to its present location, away from the southeastern branch of the Mid-Indian ridge system, during the Tertiary crustal-spreading episode.

No evidence is known to suggest that the island is of continental origin. Alkali syenite bodies, some of which are rich in quartz, were intruded into shield basalt in the southwestern part of the island during late Miocene to early Pliocene time, and are late-stage differentiates, similar to those found on other oceanic islands. Even though direct sampling of unexposed parts of the ridge (during the Deep-Sea Drilling Project) may yield lower Tertiary to upper Mesozoic sedimentary and igneous rocks, it is considered unlikely that any of the samples will consist of continental material.

INTRODUCTION

Dietz and Holden (1971) emphasized that there may be sialic crust in the southern Indian Ocean, in the form of the major aseismic features of Broken Ridge and the Kerguelen-Gaussberg ridge (Fig. 1). In their analysis of crustal genesis in the Indian Ocean, they assumed that these two features were continental fragments. If these aseismic ridges are in fact microcontinental, then the island of Kerguelen might be expected to exhibit properties atypical of oceanic islands.

Assuming that such an island has no obvious continental sedimentary and fossiliferous rocks, a true microcontinental fragment could nevertheless be composed of sialic igneous or metamorphic rocks of an age much greater than that of the adjacent sea floor, such as is found in the case of the Seychelle Islands (Baker and Miller, 1963). Sr-isotopic ratios in igneous rocks may be consistent with remobilization of deep sialic material as in Rockall (Moorbath and Welke, 1968; Scrutton, 1972), and paleomagnetic data may indicate a lack of lateral crustal mobility. Basalt geochemistry is rarely diagnostic of oceanic or continental origin, as although abyssal basalt is distinct from any basalt of continental affinity, the basalt of oceanic islands, even if tholeiitic, tends to be richer in the potassium-group elements and similar in most respects to some flood basalt of the continents. Geophysically observed subsurface properties such as seismic velocities and densities can also contribute to diagnosis of microcontinental material, while direct sampling by deep-drilling can perhaps provide the most convincing evidence.

PREVIOUS GEOLOGICAL WORK

Despite the fact that Kerguelen is the world's second largest mid-oceanic island (after Iceland), few geological studies have been made.

The Kerguelen archipelago consists of about 400 islands covering a land area of about 5,000 km². The main island of Kerguelen (Fig. 2) is about 160 km wide and is built mainly of overlapping basaltic shield volcanoes of Hawaiian type which together with some minor intrusive rocks totals some 6,000 km³ in volume for that part of the island which is above sea level. Kerguelen has been deeply dissected by glacial action, leaving a coastline indented by fiords.

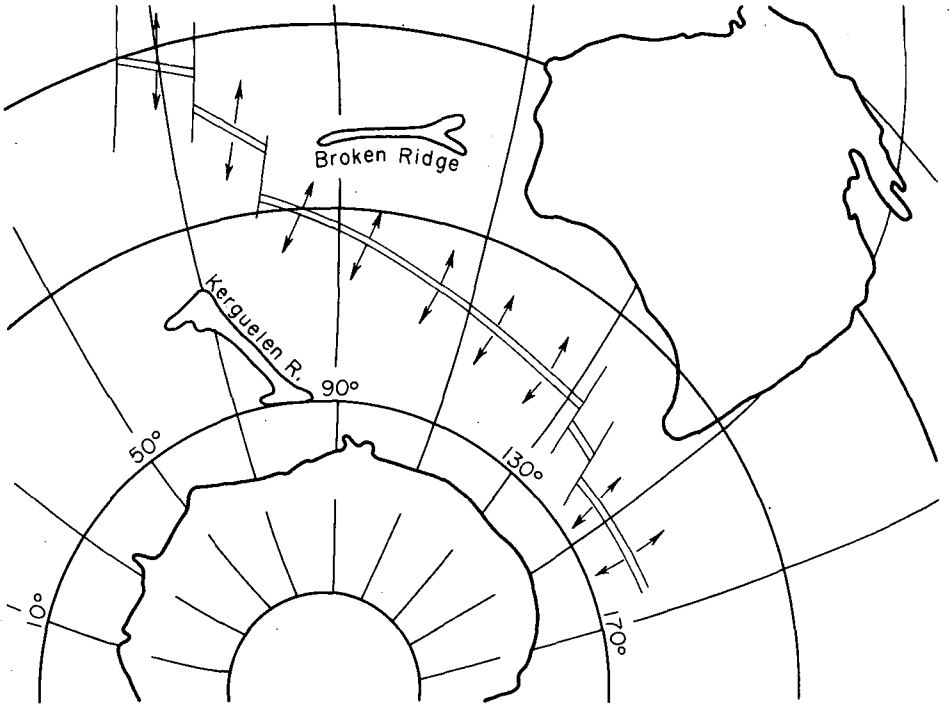


Figure 1. Sketch diagram showing the preferred conjugate relation between Kerguelen Ridge and Broken Ridge, flanking the spreading center of the mid-oceanic southeast Indian Ridge (from Dietz and Holden, 1971).

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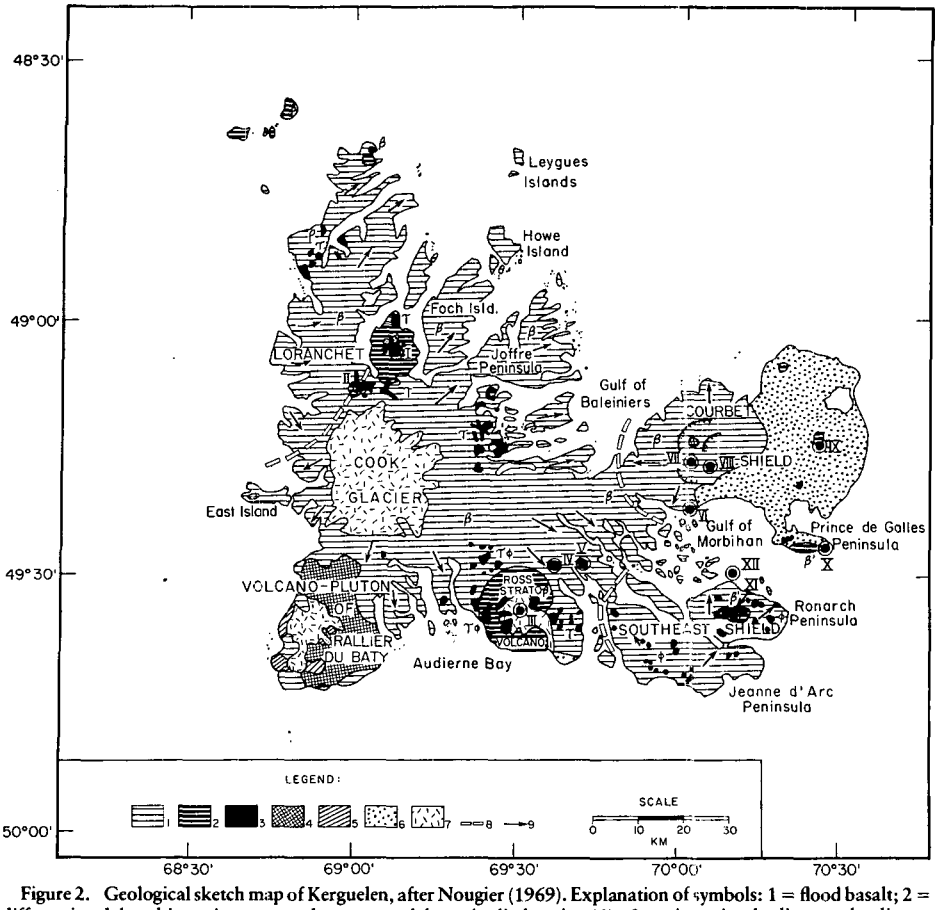


Figure 2. Geological sketch map of Kerguelen, after Nougier (1969). Explanation of symbols: 1 = flood basalt; 2 = differentiated basaltic series–stratovolcanoes, and basanite-limburgite (β); 3 = intrusive bodies — rhyolite to quartziferous trachyte (ρ), alkali trachyte (τ), nepheline trachyte (τ-Φ), and phonolite (Φ); 4 = plutonic body; 5 = Holocene trachyte flows and lahars associated with plutonic rocks; 6 = moraines; 7 = ice cap and glaciers; 8 = suspected basalt flow limits; 9 = regional dip of extrusive rocks. Geographic localities are: (i) = Geographic Society Volcano, (ii) = Guynemer Volcano, (iii) = Buffon Glacier, (iv) = Puy St. Theodule, (v) = Diable Volcano, (vi) = Cape Kidder, (vii) = Val Studer, (viii) = Mica Ravine, (ix) = Mount Peeper, (x) = Pointe Suzanne, (xi) = Mount Thompson, and (xii) = Isle Suhm.

Glaciers persist to the present day, and outlet glaciers from the Cook Glacier Cap calve icebergs into the sea on the western coast.

Nougier (1969) showed that the island is formed by the overlapping of four main shield volcanoes on which are superimposed a series of younger alkaline stratovolcanoes, adventive cinder cones and basanitic domes, phonolite plugs, trachyte domes, and rhyolite flows. The main structural units are given below. The petrological descriptions of some of the units result from work reported in this paper.

Shield Volcanoes

1. The Northwestern, L'Oranchet Shield Volcano. This center is deeply dissected and eroded. The flows appear to originate from the west coast so that much of this center has been reduced by marine erosion to below sea level. The flows are transitional in composition from alkaline to tholeiitic basalt and include rhyolite flows emanating from a series of minor centers.

2. Cook Shield. The basalt flows radiate from a center near the Cook Ice Cap, extending as far as the Joffre Peninsula 60 km to the northeast. Minor trachyte dikes and flows are found. The flows appear to be mainly tholeiitic in composition.

3. Jeanne D'Arc Shield Volcano. This center has also been much reduced by erosion. All flows dip to the north from a point off the southern coast. The Mount Thomson basanite dome is superimposed on the tholeiitic shield basalt flows on the Ronarch Peninsula, and numerous phonolite plugs are found both here and on the Jeanne D'Arc peninsula.

4. De Courbet Shield. This is apparently the youngest and best preserved of the old shield volcanoes and is surmounted by an older caldera 12 km across and a younger neo-caldera. The remnants of the rim show it to be about 6 km wide. The rocks are olivine basalt and (or) hawaiite.

Stratovolcanoes

Several massive stratovolcanoes superimposed on the shield basalt show comparatively minor glacial modification and may have been active until mid-Pleistocene time. These include:

1. Ross Volcano. Situated on the south coast, this volcano is 10 to 12 km across and 1,850 m high. The crater wall has been breached by the Buffon glacier which flows from it to the sea. Trachyte flows are interspersed with the dominant alkali-basalt flows, and a series of parasitic trachyte domes occur on the flanks of the main cone.

2. Geographic Society Volcano and the Smaller Guynemer Volcano. Situated on the boundary between the Cook and the L'Oranchet shields (1,940 and 100 m high, respectively), these volcanoes are, like the Ross Volcano, formed of alkali-basalt flows and breccia with minor trachyte.

Minor Adventive Cones

Small alkali basalt adventitious cones occur on the flanks of the old shield volcanoes and are similar to or younger in age than the Ross Volcano. These include Mount Peeper which rises out of the fluvioglacial plains in the east, the Volcan du Diable, and Puy St. Theodule (Fig. 2).

Basanite Domes

Two prominent basanite domes occur on the southeast of the island, the dome of Mount Thomson on the Ronarch Peninsula and the dome on the Prince de Galles Peninsula on the opposite side of the Passe Royale. Basanite dikes occur at the Mica Ravine in Val Studer within the de Courbet Shield.

Phonolite Plugs

Small plugs of phonolite occur on the Ronarch Peninsula and the Jeanne D'Arc peninsula in close association with the basanite rocks.

Trachyte Plugs and Flows

The trachyte masses occur in a broad belt oriented southwest to northeast. They occur as intercalated flows, dikes, and as parasitic domes on the stratovolcanoes in the older shield basalt and in the Rallier du Baty complex. The trachyte includes quartz-free alkali trachyte as well as quartz-bearing varieties.

Rhyolitic Flows

Rhyolite is limited to a few flows in the Loranchet shield in the northwest.

Plutonic Complex

The Rallier du Baty complex is in the extreme southwest part of Kerguelen (Fig. 2). This is an intrusive complex of gabbro,

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diorite, syenite, nordmarkite, and syenite (Nougier, 1963, 1966, 1969). Extensive trachyte flows are associated with the complex as well as more feldspathic rocks with andesine plagioclase which have been called trachybasalt, although hawaiite may be a more appropriate term. Nougier and Lameyre (1972) draw attention to the mineralogical and geochemical similarity of the pluton to bodies in tectonically active parts of continents.

Sediments

The only lithified sediments found in Kerguelen tend toward lignite containing araucarites, the ages of which are suspected to be mid-Miocene. No organic limestone, such as the suspected lower Tertiary outcrops on Heard Island (at the southern end of the Kerguelen plateau) described by Lambeth (1952) and Stephenson (1964), are found on Kerguelen.

PREVIOUS GEOPHYSICAL WORK

Schlich and Patriat (1971) have analyzed marine magnetic anomaly patterns between Kerguelen and Amsterdam (Fig. 3) and conclude that anomaly 11 (t = 35 m.y.) is the oldest marine magnetic anomaly which can be identified north of Kerguelen (Fig. 4). LePichon and Heirtzler (1968) believe that anomaly 17 (t = 43 m.y.) can be identified southeast of Kerguelen. These data provide minimum ages for the sea floor adjacent to Kerguelen.

McKenzie and Sclater (1971) derived a comprehensive history of the formation of the Indian Ocean since early Mesozoic time, based largely on analyses of marine magnetic anomalies. Their preferred reconstruction involves (as suggested by Dietz and Holden, 1971) adjacent alignment of the eastern side of the Kerguelen-Gaussberg Ridge and the southern edge of Broken Ridge (Fig. 1), at about t = 45 m.y. It thus appears that t = 35 to 45 m.y. would be the approximate minimum age for any pre-existing continental fragment constituting the Kerguelen plateau.

Schlich and others (1971) published some seismic reflection profiles taken on the flanks of the Kerguelen plateau south and southeast of Kerguelen. They interpret the data to indicate steep faulting as the cause of the elevation of the plateau above the sea floor, and they favor a microcontinental rather than an oceanic origin for the feature.

Ewing and others (1972) have presented sonobuoy seismic refraction lines taken across the plateaus. The highest velocities encountered at depth are not mantle velocities but are more consistent with consolidated older sediment beneath the plateaus.

Irving and others (1965) made a paleomagnetic analysis on a limited collection of basalt samples of Pliocene-Pleistocene age from Heard Island (Stephenson, 1972). Data (mean virtual geomagnetic pole of 79° N., 45° E., α_{95} = 9°) are interpreted to represent a possible increase in the latitude of Heard Island (and the Kerguelen Plateau), consistent with sea-floor spreading southward from the mid-oceanic ridge system to the north (Fig. 1). Nougier and Plessard (1968) have obtained, however, a mean virtual geomagnetic pole of 76° N., 206° E. for seven lava flows of early Miocene age (according to a commercially obtained K:Ar age) on the southwest part of the Courbet peninsula of Kerguelen. The available paleomagnetic data would appear to be from collections which are too limited in size to resolve independently the possibility of crustal movement of the Kerguelen Plateau, consistent with migration from the north. Paleomagnetic data could possibly distinguish between the proposed southward migration of the Kerguelen Plateau away from the oceanic ridge system and an alternative northward migration of the spreading ridge.

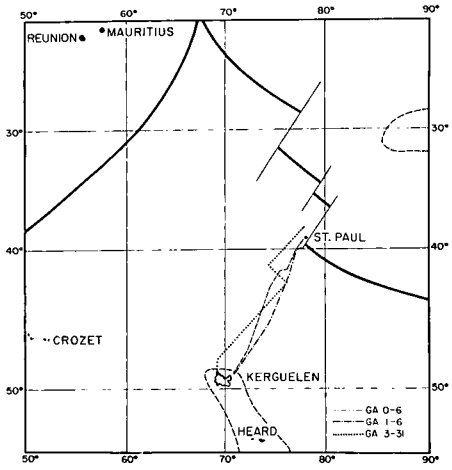


Figure 3. Sketch diagram showing location of three magnetic profiles (GA0-6, GA1-6, and GA3-31) between St. Paul Island and Kerguelen. Approximate location of mid-oceanic ridge system shown. Kerguelen-Gaussberg Ridge outlined. From Schlich and Patriat (1971).

PREVIOUS WORK IN ISOTOPE GEOCHEMISTRY

From our summary of the previous work, it is clear that isotopic measurements are potentially very significant contributors to

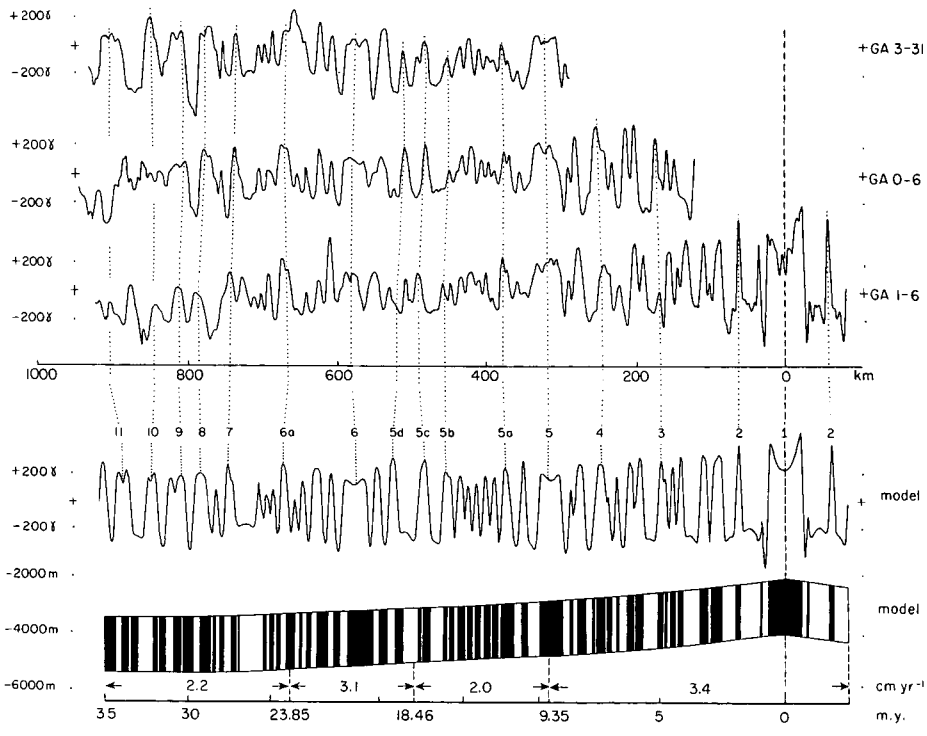


Figure 4. Crustal-spreading analysis of marine magnetic profiles. Three profiles shown at top of diagram (profile numbers correspond to Fig. 3). Model and corresponding anomaly pattern computed using reversal time scale of Heirtzler and others (1968). Anomaly numbers also due to Heirtzler and others (1968). Interpreted age of anomaly in m.y. at base of diagram. Interpreted crustal spreading rates (in cm yr^{-1}) shown. Model depths and thickness according to depth scale (in m) on left. Amplitude of observed and model anomalies shown in gammas on left. From Schlich and Patriat (1971).

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resolving the problem of Kerguelen's proposed continental origin.

As discussed above, an age in excess of about 50 m.y. could be interpreted to indicate that the island is older than the adjacent sea floor. This would allow a continental origin.

No Sr-isotope determinations have been reported for Kerguelen rocks. McKenzie and Sclater (1971) have recently drawn specific attention to the desirability of Sr-isotope studies of the Rallier du Baty plutonic body in order to distinguish between an origin from remobilized basement and one from shallow differentiation. They refer to it as a granitic body (Bellair and Nougier, 1963), but we think that it is more correctly described as being syenitic in composition.

Nougier (1969) has compiled a series of commercially obtained K-Ar ages which show that alkali basalt dates from Miocene through Pliocene, and Molchanova (1972) has recently presented four K-Ar dates of essexite samples collected by Soviet geophysicists during an investigation of the Courbet peninsula in 1966. She describes the rocks as being strongly altered (olivines completely replaced), and the measured ages, with a mean of 34.5 ± 2.0 m.y., must therefore be suspect.

No Rb-Sr age estimates have yet been obtained.

PREVIOUS WORK IN GEOCHEMISTRY

Published geochemical data for Kerguelen and Heard Island are sparse. Stephenson (1972) interprets the major-element geochemistry of a suite of igneous rocks from Heard Island in terms of crystal fractionation, which is similar to Edward's (1938) interpretation of data from a few Kerguelen samples. Molchanova (1972) has published analyses of 13 specimens from the Courbet Shield. While no stratigraphic control was obtained, she concludes that the results are consistent with differentiation processes beginning with crystallization of oceanic tholeiitic basalt, followed by increasing alkalinity and acidity with differentiation, to alkalic microgranite and phonolite trachyte (by way of quartz microsyenite) as the extreme members, as compared to continental granite.

Nougier (1969) analyzed a series of olivine and tholeiitic basalt flows from Kerguelen. The SiO_2 content is about 48 percent; the average Na/K ratio is 3; and the U and Th averages (Cheminee and Nougier, 1972) are higher in the olivine basalt. These flows are interpreted to be evolving toward hawaiite and mugearite.

PURPOSE OF PRESENT STUDY

We shall present the results of geochemical, isotopic, and paleomagnetic analyses of samples of igneous rocks from selected parts of Kerguelen. Nougier (1969) suggested that the basalt is monotonously uniform in character with few accumulative rocks either of oceanite or ankaramite; nevertheless, transitional tholeiitic and mildly alkaline lava flows are both present. A total of 165 Kerguelen rocks were analyzed by Nougier (1969) for the major elements, and although the analytical precision for some elements, especially K and Al, was low, it appeared that among the shield volcanoes, some flows (especially those of the central Cook shield volcano, and of one of its prolongations, Foch Island) are mainly quartz normative and carry characteristically low K and TiO_2, compared with the more alkaline, often nepheline normative rocks of the de Courbet shields and younger centers. Accordingly, our detailed study has been made on the following sequences.

1. A series of tholeiitic basalt flows at Port Mary, at the foot of Mount Bureau, Foch Island (Fig. 5).
2. Three sequences of basalt flows on the de Courbet peninsula (Fig. 5): at Cape Kidder on the coast on the flanks of the volcano, Mount du Chateau, and La Grande Cascade within the youngest caldera.
3. A series of basanite and alkali basalt flows in the Prince de Galles peninsula; from outcrops in Val Studer, which bisects the de Courbet cone; and from Mount Peeper (Fig. 2).

The unpublished geochemical data of Nougier (1969) has also been re-examined.

Paleomagnetic studies have been made of the sections from Foch Island (Fig. 5) and the three sections on the de Courbet peninsula. Detailed K-Ar analyses have been made of tholeiite from Foch Island.

Finally, we shall report the results of Rb-Sr age estimates and Sr-isotopic analyses of samples from the Rallier du Baty pluton (Fig. 6), and a single gabbroic intrusion from the de Courbet peninsula.

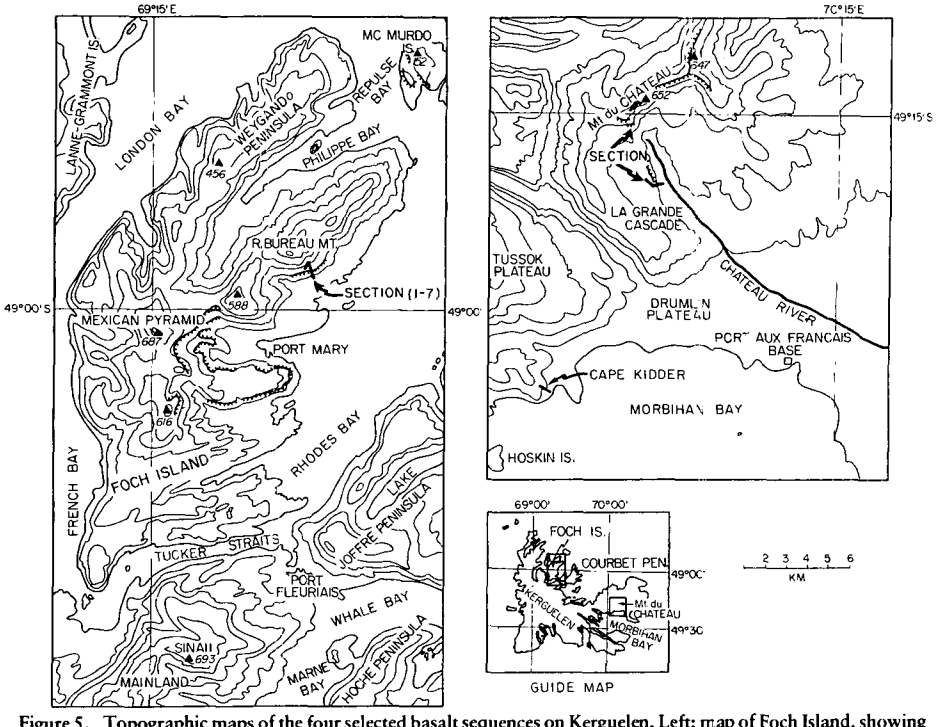


Figure 5. Topographic maps of the four selected basalt sequences on Kerguelen. Left: map of Foch Island, showing location of Mount Bureau section. Right upper: map of south-central part of Courbet peninsula, showing location of Mount du Chateau, Grande Cascade, and Cape Kidder sections. Right lower: location of maps. Contour interval is 100 m. Topographic spot heights in meters.

METHODS

Field

Samples were taken from the four basalt sequences using a gasoline-powered portable drill. A total of 128 cores 2.5 cm in diameter and with an average length of 15 cm were obtained. All cores were oriented in geographic coordinates. Hand samples were taken from the basanite outcrops, and from the Rallier du Baty pluton. Helicopter transport was used throughout.

Laboratory

Geochemistry. Nougier (1969) described the conventional wet methods used in his analyses, which have been presented elsewhere, but which we shall incorporate here.

The methods used for the new results which we shall present have been described in detail by Gunn and Watkins (1969). Major and minor elements are determined by x-ray fluorescence spectrometry with the exception of soda which was done by atomic absorption. Ferric iron was not determined separately; a constant amount of 2 percent was allotted in the norm calculations.

The results add up to 98 to 99 percent

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exclusive of volatiles and have been summed up to 100 percent for greater comparative accuracy. In several instances, 2 or 3 drill cores per flow were analyzed, yielding a measure both of analytical precision and sample homogeneity. All statistics and data plots were made by CDC 7400 computer. All samples were examined petrographically prior to analysis.

Paleomagnetism. The remanent magnetism of 2.2-cm-length specimens from each core was measured using a 5 herz spinner magnetometer, with maximum sensitivity sufficient to measure directions to within ±5° for a total moment of 8×10^{-7} emu. Alternating magnetic fields of 100, 200, and 300 oersteds were applied to each specimen, following remeasurement of the remanent magnetism direction and intensity after each treatment. Mean directions of remanent magnetism were computed using the minimum dispersion criteria (Watkins and others, 1972).

K-Ar Ages. Samples were crushed to 10 to 30 mesh size (Baksi, 1972), for the argon determinations. A portion was further crushed to −100 mesh size, for the potassium analyses. Samples were loaded in pre-fired niobium crucibles, and fusion was achieved by radio-frequency induction heating. The gases were spiked, purified, and then analyzed on an MS10 mass spectrometer, operated in the static mode. Where necessary, the appropriate orifice correction (Baksi and Farrar, 1974) was applied.

No discrimination correction was applied to the measured isotopic ratios, as during the course of this work, the Ar^{40}/Ar^{36} ratio for atmospheric argon samples always fell in the range 295.4 to 296.4.

Potassium analyses were carried out using an I.L. 143 flame photometer. Details of the method used are available elsewhere (Baksi and others, 1967). For each sample, at least two analyses were carried out. In no case did the individual measurements differ by more than 3 percent. The error limits were computed using the method of Cox and Dalrymple (1967).

Rb-Sr Ages and Strontium Isotopes. The strontium-isotopic and rubidium analyses for the plutonic specimens were kindly made by C. E. Hedge, U.S. Geological Survey, using the procedures given in Hedge and Peterman (1970).

RESULTS AND DISCUSSION

Local Geology and Petrography

Foch Island. The Mount Bureau section (Fig. 7) is a massif 600 m high consisting of a gently northwesterly dipping basal unit underlying lava flows 200 to 250 m thick. The seven sampled flows (KG1 to KG7) have a total thickness of about 150 m and are from the lower series.

The tholeiitic basalt flows are holocrystalline but invariably fine grained. Micro-

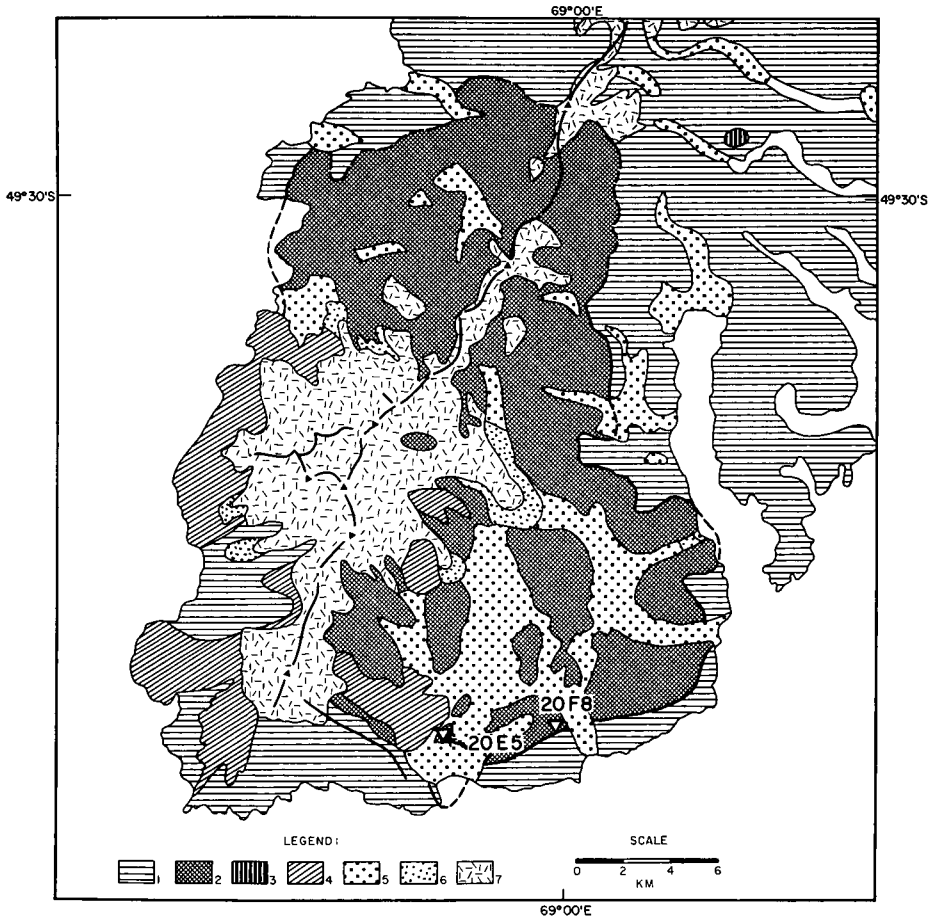


Figure 6. Geologic map of Rallier du Baty peninsula. Explanation of symbols: 1 = plateau basalt; 2 = Rallier du Baty pluton; 3 = trachytic flows independent of the intrusion; 4 = trachytic flows and lahars associated with the pluton; 5 = fluvioglacial material; 6 = moraines; 7 = glacier. Topographic divides marked by heavy line. From Nougier (1969). Locations of samples used for Rb and Sr isotopic analyses (Table 3) added.

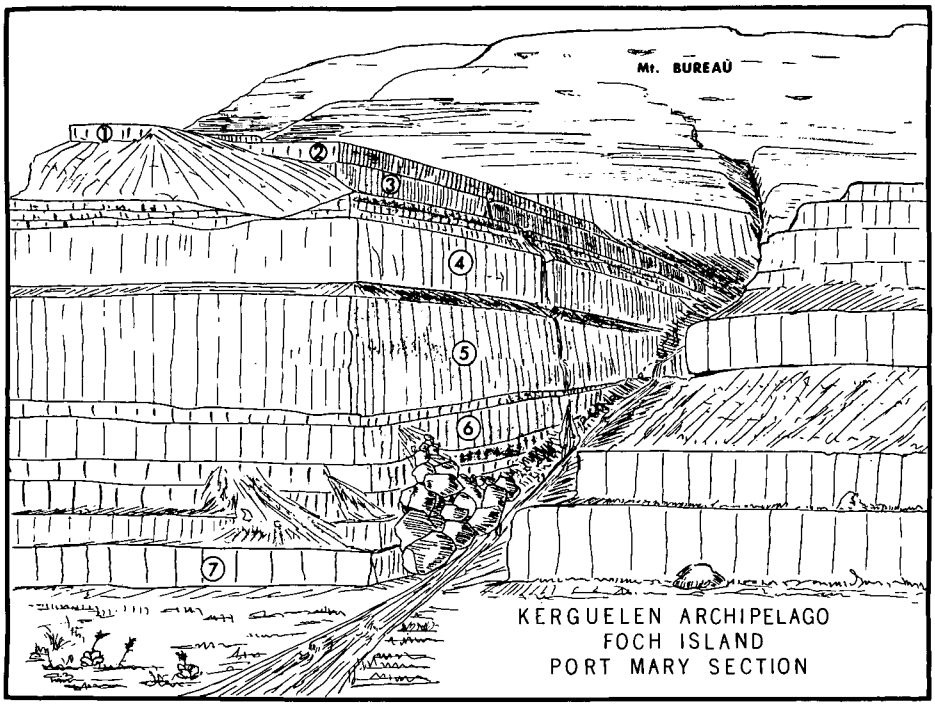


Figure 7. Sketch of the Mount Bureau section (Fig. 5). Numbers on flows correspond to chemical and petrographic analyses (Tables 1 and 2), paleomagnetic results (Table 5) and K-Ar data (Table 3).

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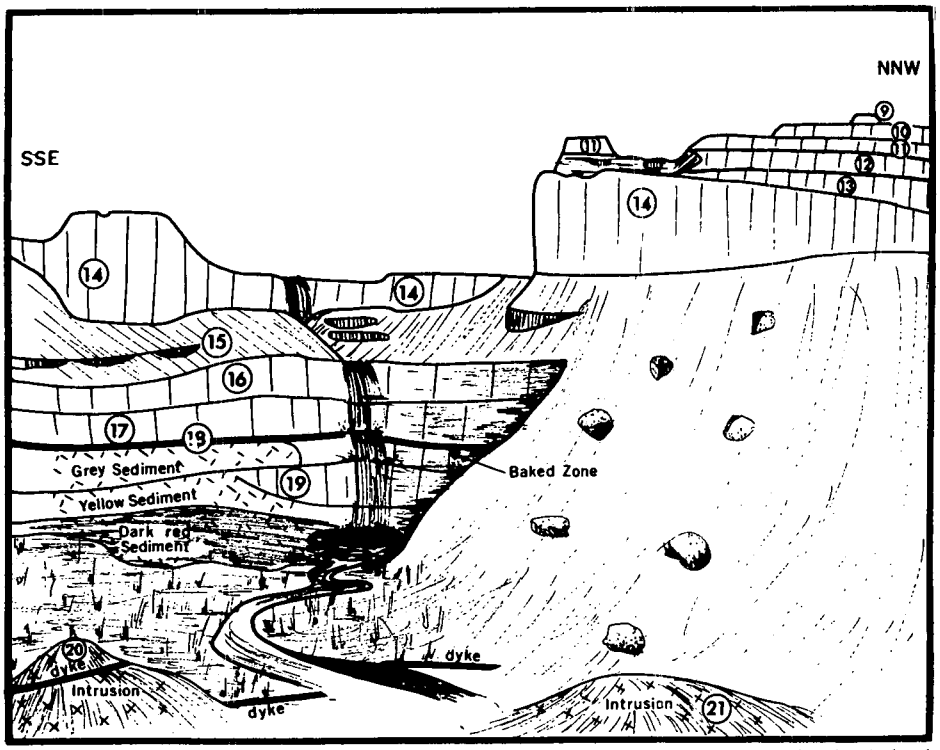


Figure 8. Sketch of the Kerguelen Grande Cascade section (Fig. 5). Numbers on bodies correspond to chemical and petrographic analyses (Tables 1 and 2), paleomagnetic results (Table 5), and K-Ar results (Table 3).

phenocrysts, seldom exceeding 1 mm, are present, rarely of olivine (KG4-1: 4th lava flow, 1st core), more often of plagioclase together with olivine (KG4-2). The groundmass consists of oriented laths of plagioclase and granular clinopyroxene and titanomagnetite. Abundant chlorophaeite may be present (KG3-1, KG2-3), but the rocks are too fine grained to distinguish the residuum. In one specimen (KG-1), the clinopyroxene appears as granular patches 1 to 2 mm across, with a common orientation, but no phenocrysts of olivine and clinopyroxene appear together, as is usually seen in alkali basalt.

Courbet Peninsula. The "Grande Cascade" section is about 7 km southeast of the center of the Courbet shield (Fig. 5). It consists (Fig. 8) of 12 lava flows overlying a shallow intrusive trachyte. The flows are olivine basalt with abundant clinopyroxene and olivine phenocrysts 1 to 2 mm wide (KG8-1, 9-1, 10-4, 11-2). Plagioclase is always prominent; in hawaiite, it forms the dominant phenocryst phase (KG15-1). In hawaiite, olivine and clinopyroxene phenocrysts may be present or absent depending on the degree of fractionation. Zeolite is often prominent in amygdales and scattered through the groundmass (KG15-4).

There is little distinction between the olivine and tholeiitic basalt, except that the dual appearance of clinopyroxene and olivine phenocrysts appears to be confined to the former. The residual products of fractionation are the hawaiite in the olivine basalt series and a plagioclase-rich rock significantly lower in K and Rb and Ti, compared to its alkaline equivalent, in the tholeiitic series, for which no name has been put forward, although Icelandite would be close to the same composition, if a slightly higher SiO_2 concentration was evident.

Basanite. A detailed petrographic study of the basanite of Pointe Suzanne has been made by Jeremine and Sandrea (1958). Some basanite at Mount Thomson in presqu'ile Ronac'h, together with limburgite and leucitic tephrite, have been described by Edwards (1938).

The massif of the presqu'ile Prince de Galles is a single eruptive center. The exposed section shows thick columnar flows, built of analcimic limburgite and of basanite with nodules of olivine. At Pointe Suzanne, lava flows are analcimic basanite with small phenocrysts of titaniferous clinopyroxene and olivine, which also occur in the brown vitreous groundmass as prisms and laths, associated with grains of titanomagnetite. The analcime shows anhedral crystals pitted with concentrically arranged inclusions. Feldspar phenocrysts are commonly subordinate to the numerous grains of titanomagnetite.

On the Courbet peninsula, there are several small basanitic volcanoes. Mount Peeper is a composite cone of basanitic lava flows and plugs. The flows are porphyritic, with phenocrysts of olivine, pyroxene, and titanomagnetite.

Rallier du Baty Pluton. A series of syenitic plutons covers much of the Rallier du Baty peninsula (Fig. 6). The older shield basalt flows have been domed up, stoped, block faulted, and retrograde metamorphosed by the intruding syenite masses. Nougier and Lameyre (1972) have pointed out that these rocks are frequently quartz bearing, and some have granitic affinities. Other small syenitic stocks are found elsewhere on the island, notably at Mont Vert in the de Courbet shield and on the Ile de Ouest. The petrography of the different stocks is extremely varied. The most basic varieties (monzonite) have plagioclase dominant over potash feldspar and contain a titaniferous clinopyroxene often together with biotite. Nepheline syenite contains coarse patch-perthite, a sodic amphibole, and often a dark blue-green riebeckite. Quartz-bearing varieties have either granular quartz, perthite, and green-brown hornblende with a granitoid texture or have a granophyric texture (sample 20F8 in Fig. 6). Though more than 20 samples of the plutonic rocks have been chemically analyzed, sufficient field work has not yet been done to establish the range of variation within a single apophysis of the complex. The compositions of the quartz-bearing and nepheline-bearing syenites parallel the quartz and nepheline trachytes.

Petrochemistry

A petrochemical progression from quartz-tholeiite to basanite-phonolite is present in the extrusive rocks of Kerguelen. In Table 1, we present average analyses for different rock groups constituting this series, from all available data, including the new analyses. In Table 2, we present the trace-element data, resulting from the new analyses.

Courbet Peninsula Olivine Basalt. Our analyzed samples were collected from 19 flows within the neocaldera, 6 flows from the base of the cone at Cape Kidder, and 2 from Val Studer, a deep glacial valley which almost bisects the cone. Of these, 8 flows were analyzed in duplicate or triplicate using separate drill cores. Previous experience has shown that the residual elements may be localized within the upper parts of flows, associated with fractionation of K-Rb, K-Ba and K-Cs values (Watkins and others, 1970; Hart and others, 1971). For the Courbet peninsula flows, the maximum relative variation in K_2O was found to be ±7 percent, for a single flow 8 m thick. The general thinness of the flows appears to be related to minimal intraflow variations. One

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TABLE 1. AVERAGE CHEMICAL COMPOSITIONS (MAJOR ELEMENTS) AND CIPW NORMS OF SELECTED GROUPS OF KERGUELEN IGNEOUS ROCKS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
SiO_2	49.00	49.95	49.31	49.44	48.18	50.80	48.84
Al_2O_3	15.46	19.48	16.79	15.65	15.64	15.82	15.12
TiO_2	2.12	0.60	1.40	1.21	2.89	2.95	2.95
Fe_2O_3	12.49	7.94	7.51	8.35	12.91	12.84	14.03
FeO	0.00	4.15	5.51	4.49	0.00	0.00	0.00
MnO	0.17	0.19	0.33	0.17	0.17	0.17	0.18
MgO	7.14	4.83	5.52	4.84	6.18	3.75	5.19
CaO	10.36	8.44	8.69	10.76	8.87	7.99	9.19
Na_2O	2.54	3.18	3.08	3.13	3.18	3.70	2.76
K_2O	0.48	0.68	1.57	1.32	1.65	1.58	1.36
P_2O_5	0.23	0.66	0.66	0.64	0.46	0.48	0.36
Quartz	0.00	5.30	1.78	0.00	0.00	0.00	0.00
Orthoclase	2.86	4.00	9.28	7.82	9.74	9.33	8.02
Plagioclase	50.85	63.79	53.42	51.23	50.44	53.21	48.22
Nepheline	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Diopside	16.92	0.87	9.39	19.36	14.58	12.45	15.43
Hypersthene	17.88	12.05	11.51	7.75	0.34	13.31	14.62
Olivine	3.01	0.00	0.00	0.23	14.64	1.22	3.19
Magnetite	2.90	11.51	10.89	11.53	2.90	2.90	2.90
Ilmenite	4.02	1.14	2.68	2.29	5.27	5.45	5.61
Hematite	0.00	0.00	0.00	0.40	0.00	0.00	0.00
Apatite	0.50	1.44	1.44	1.40	1.00	1.05	0.79

	(8)	(9)	(10)	(11)	(12)	(13)	(14)
SiO_2	47.27	48.03	45.75	45.10	72.08	52.23	62.26
Al_2O_3	12.45	17.78	12.45	16.42	14.76	21.19	17.63
TiO_2	3.30	1.61	3.26	1.53	0.30	0.37	0.18
Fe_2O_3	9.07	6.84	10.93	5.41	1.63	5.56	3.17
FeO	0.00	5.13	0.00	5.63	1.38	1.90	2.31
MnO	0.21	0.15	0.21	0.26	0.05	0.12	0.13
MgO	11.81	4.23	10.75	6.86	0.46	4.19	1.13
CaO	10.31	9.24	9.85	11.13	0.74	7.21	1.60
Na_2O	2.71	3.74	3.53	3.88	4.15	4.25	6.39
K_2O	2.09	2.32	2.20	2.75	4.96	2.44	5.74
P_2O_5	0.79	0.92	1.06	1.04	0.27	0.52	0.21
Quartz	0.00	0.00	0.00	0.00	27.02	0.00	0.00
Orthoclase	12.33	13.72	13.03	16.25	29.29	14.41	33.94
Plagioclase	31.65	52.46	23.71	26.00	37.20	67.51	55.39
Nepheline	3.75	2.20	9.63	14.14	0.00	0.00	0.62
Diopside	24.42	12.27	24.80	23.72	0.00	0.91	3.44
Hypersthene	0.00	0.00	0.00	0.00	1.93	1.15	0.00
Olivine	16.25	4.36	16.52	6.89	0.00	7.86	1.97
Magnetite	2.90	9.92	2.90	7.84	2.36	5.43	4.60
Ilmenite	6.27	3.07	6.20	2.91	0.57	0.71	0.34
Hematite	0.00	0.00	0.00	0.00	0.00	1.81	0.00
Apatite	1.72	2.01	2.31	2.27	0.59	1.13	0.46

	(15)	(16)	(17)	(18)	(19)	(20)	(21)
SiO_2	59.48	59.73	62.20	63.19	57.75	60.67	56.95
Al_2O_3	18.95	20.59	18.51	17.40	17.68	19.24	20.42
TiO_2	0.37	0.14	0.30	0.32	0.59	0.24	0.29
Fe_2O_3	3.54	3.80	3.18	2.51	5.04	3.49	2.77
FeO	2.29	1.21	2.16	1.53	2.03	1.86	1.90
MnO	0.16	0.15	0.16	0.09	0.15	0.15	0.15
MgO	0.84	1.33	0.88	0.84	3.81	1.08	1.05
CaO	2.30	1.83	1.42	0.97	3.58	1.82	1.76
Na_2O	6.00	6.43	6.00	5.64	5.28	6.21	7.84
K_2O	5.99	5.12	5.37	5.31	3.68	5.48	6.92
P_2O_5	0.38	0.42	0.33	0.23	0.40	0.36	0.23
Quartz	0.00	0.00	2.71	7.14	0.00	0.00	
Orthoclase	35.44	30.26	31.74	31.38	21.77	32.39	40.88
Plagioclase	51.40	57.45	55.88	51.19	58.34	58.78	25.98
Nepheline	3.49	1.93	0.00	0.00	0.00	0.37	21.92
Diopside	1.71	0.00	0.00	0.00	1.36	0.00	5.85
Hypersthene	0.00	0.00	3.33	2.47	9.82	0.00	
Olivine	1.61	3.53	0.00	0.00	0.94	2.20	0.57
Magnetite	5.13	3.99	4.61	3.64	5.31	5.06	4.02
Ilmenite	0.71	0.27	0.57	0.61	1.12	0.46	0.56
Hematite	0.00	1.05	0.00		1.37	0.00	
Apatite	0.83	0.92	0.72	0.90	0.87	0.79	0.50

Major elements are in weight percent; trace elements are in parts per million. Compositions of groups are: 1 = 9 tholeiitic basalts, Foch Island; 2 = 27 basalts, Cook Shield; 3 = 17 olivine basalts, Jean d'Arc Peninsula; 4 = 6 olivine basalts, Loranchet peninsula; 5 = 15 olivine basalts, Mount du Chateau; 6 = 6 hawaiites, Mount du Chateau; 7 = 9 olivine basalts, Cape Kidder; 8 = 7 alkali basalts, Mount du Chaleen and Mount Peeper; 9 = 5 basalts, adventitious cones; 10 = 14 basanites, Prince de Galles Peninsula; 11 = 9 basanites, Mount Thompson; 12 = 13 rhyolites, Loranchet peninsula; 13 = 7 trachyandesites (hawaiites) from Rallier du Baty Peninsula; 14 = 9 trachytes, Courbet Peninsula; 15 = 14 trachytes, southeast shield; 16 = 22 trachytes, Mount Ross; 17 = 15 trachytes, Rallier du Baty; 18 = 19 syenites, Rallier du Baty pluton; 19 = 1 trachyte, Loranchet Shield; 20 = 60 trachytes, representing entire Kerguelen collection; 21 = 24 phonolites, Mount Thompson area. Source of data: groups 1, 5, 6, 7, 8, and 10 this paper; other groups Nougier (1969). Note that the Nougier data tends to be high in Al_2O_3 and low in CaO and MgO because of the methods used (see text). For geographic sites, see Figure 1 and text. For experimental methods, see text.

of the flows (KG13-1) has markedly lower K_2O (0.42 percent compared to an average of 1.9 percent) and lower Ti, P, Sr, Rb, Ba; it is quartz tholeiite included within a predominantly olivine-tholeiite basalt pile. The rest of the flows lie between the Di-Ol-Ab and the Di-Ab-Hy planes of the basalt tetrahedron, though some of the more highly fractionated rocks have one or two percent of normative nepheline. Lava from within the caldera has a higher mean alkali content and lower Fe + Mg than lava from the flank of the volcano, but is similar otherwise.

Despite the lack of accumulative oceanite or ankaramite, there is a wide range in element abundances. Of the ferromagnesian elements, MgO has a range of 3 to 9 percent,

TABLE 2. AVERAGE CHEMICAL COMPOSITIONS (TRACE ELEMENTS) OF SELECTED GROUPS OF KERGUELEN IGNEOUS ROCKS

	(1)	(5)	(6)	(7)	(8)	(10)
Cr	143.4	108.6	16.5	67.9	..	..
Ni	99.9	94.3	27.0	56.7	..	..
Cu	81.8	68.0	46.0	93.2	..	..
Rb	6.4	34.3	48.6	31.1	55.9	66.7
Sr	335.8	496.9	459.0	413.0	786.8	1,081.4
Ba	..	..	352.0	262.9	723.3	651.4
Pb	1.1	2.0	..	..	..	..
Th	2.5	6.0	..	..	..	..

Concentrations are in ppm. For samples constituting group number, see Table 1.

but Ni varies sympathetically from 40 to 230 ppm, and Cr from 10 to 260 ppm. The mutual Pearson R correlation coefficients for these three elements are 0.9 or better. Total Fe remains nearly constant, but TiO_2 is enriched strongly and MnO weakly, in the more fractionated rocks.

Among the alkalies, Na and K correlate directly with Si and Al, while K is fractionated strongly with respect to Na, being more concentrated in the residual rocks, the ratio Na_2O/K_2O decreasing from 6 to 1.5. Some within-flow residual liquid migration may contribute to such extreme fractionation, but there are no other indications of such an effect, nor is it associated with the presence or absence of zeolite minerals. K and Rb have a correlation of 0.99 and a constant ratio of 377 which may be regarded as typical for oceanic alkali basalts (Fig. 9). A trachyte dike from the de Courbet volcano has the same K/Rb ratio, and in fact all the trends described above project toward trachyte composition. The K, Th relation is also constant, with a Th range of 1 to 10 ppm and a correlation coefficient of 0.8 for 15 determinations. Within the limits of analytical precision the ratio K/Th of 2,260 appears

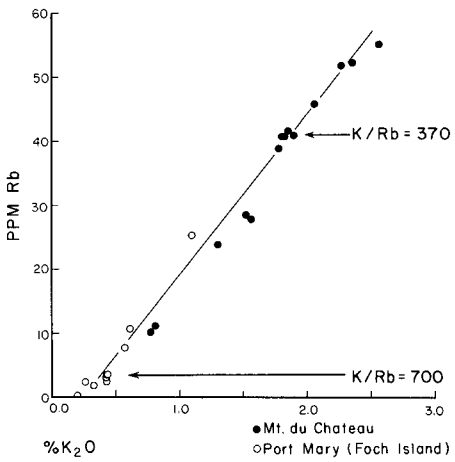


Figure 9. Relations between Rb and K_2O for the selected basalt sections (Fig. 5). In spite of the very different average K/Rb ratios of the two sections involved (an olivine basalt series at Mount du Chateau, and tholeiitic basalt at Port Mary), a single fractionated series is evident.

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to be constant. Sr has a small range of variation between 400 and 650 ppm and correlates with Na but not with Al or Ca.

Almost one-half of the rocks analyzed have normative plagioclase in the An_{40-50} range and have been provisionally termed "hawaiite." These rocks are more feldspathic and are more enriched in K, Rb, Ba, Th, Al, and Na than are the basalts. The color index of 30 to 40 to 40 to 45 for the basalt suggests that these rocks have lost Cpx + Ol prior to eruption. Mugearite and its tholeiitic equivalents are conspicuously absent.

One sample (KG21-3) is extensively replaced by a zeolite mineral, showing marked concentric zoning and perfect prismatic cleavage, which is probably phillipsite. This rock has been greatly depleted in soda.

Foch Island Tholeiite. The Mount Bureau tholeiite on Foch Island averages about 0.4 percent K_2O compared to 1.8 percent for the de Courbet olivine basalt and hawaiite (Table 1). Only 8 lava flows were sampled, but the results compare well with the 27 analyses by Nougier (1969) of the Cook shield volcano and the 8 of the Foch and Joffre Islands. The latter are systematically higher in alumina, and lower in MgO and TiO_2, possibly as the result of the gravimetric methods employed. One of the Mount Bureau lava flows (KG5-1) has higher K, Ti, P, Sr, and Rb than the mean values, but is quartz normative. Such a rock is probably a more highly fractionated tholeiite, not an interspersed alkali basalt.

In this series, Mg and Al show a +0.8 correlation instead of the −0.82 correlation seen in the de Courbet olivine basalt. This is an effect of the common olivine-plagioclase phenocryst association seen in the tholeiite. For the same reason, K_2O shows a −0.85 correlation with Al_2O_3 instead of +0.64, while no correlation is present for Al and Na, since part of the plagioclase is in the cumulative rocks, and part is residual. The Na and K covariance is similar in the two series, though the abundance of both elements is markedly less. The Mg range is similar as are the Mg, Cr, Ni relations; the Cr/Ni ratio is also constant.

The K/Rb ratio is higher at low K values in both series, reflecting the effect of plagioclase fractionation with a mean K/Rb range of 1,200 to 370 (Fig. 9). The Th level is barely above detection limits at 2 ppm, but despite some erratic Th values, the mean K/Th is close to that of the more alkaline rocks.

Courbet Peninsula Alkali Basalt. Two samples from Mount Peeper, a small cone rising from the fluvioglacial plain east of the de Courbet cone, and a single sample from l'Isle Suhm (where there is another Holocene cone) are the only alkali basalt flows analyzed. Four other samples from a prominent dike with large micaceous flakes, from the "Mica Ravine" in the hearts of the de Courbet cone, are of mixed composition varying from alkali basalt to basanite. They are probably late differentiates of the de Courbet flows. Alkali basalt, as may be expected, has higher Sr than the olivine basalt (733 ppm compared to about 450 ppm), with 400 ppm being typical for the Port Mary tholeiite. The K content of the Mount Peeper rock is abnormally low, that of l'Isle Suhm more normal at 0.6 percent, giving a K/Rb value of 360. The Mount Peeper sample is also the richest in olivine of all the samples collected.

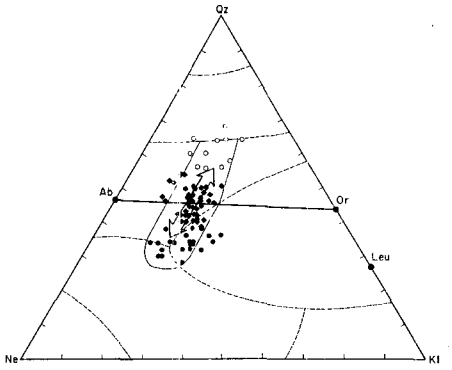


Figure 10. Quartz-nephelene-kaliophilite ternary diagram for rhyolite (○) trachyte (•), and phonolite (●) of Kerguelen (data from Nougier, 1969). The solid line encloses the low-temperature trough. The albite to orthoclase is a thermal divide. The data show that fractionation has led to both phonolite and commendite.

Basanite. These rocks also have a range of composition, almost certainly due to local concentrations of nepheline, analcite, and zeolite minerals within flows as well as to fractionation of Ol + Cpx. All rocks are rich in both olivine and nepheline with high but variable Sr, Rb, and Ba values almost double those of the olivine basalts. The mean K/Rb of 250 for seven samples from Prince de Galles Peninsula varies greatly within individual flows.

Rhyolite, Trachyte, and Phonolite. As shown in Figure 10, which is based on the data of Nougier (1969), the felsic differentiates form a continuous sequence lying along the low temperature trough in the Qz-Ne-Kl ternary diagram. As the basalt ranges in composition from alkali through transitional to tholeiitic, it spans the thermal divide at the albite-orthoclase join. Fractionation then leads toward phonolitic or commendite (Fig. 10). The Na/K ratio, though somewhat variable in the range 1 to 2, does not vary systematically with silica saturation. Trachyte has the broadest range of composition, the more basic dikes being micromonzonite. There does not appear to be a systematic regional difference in trachyte, both quartz-fayalite trachyte and alkali trachyte being found in close association. Nougier (1969) showed the same age sequence in the felsic rocks as in the basic series. However, more trachyte was erupted later, in the Ross, Geographic Society, and Guynemer volcanoes; and trachyte of Quaternary age, both as flows and tuffs, was extruded in the Rallier du Baty Peninsula.

Rallier du Baty Plutonic Complex. No new analyses were made of the intrusive rocks of the Rallier du Baty complex. The mean of 19 analyses reported by LaCroix (1924) and Nougier (1969) shows (Table 1) a syenite composition with 7 percent normative quartz, though the extremes include rocks with 10 percent nepheline and 30 percent quartz. Chemically the series shows a regular progression with alumina being in the 17 to 20 percent range for SiO_2 values between 54 and 65 percent, decreasing to 12 to 14 percent at 62 percent SiO_2, to

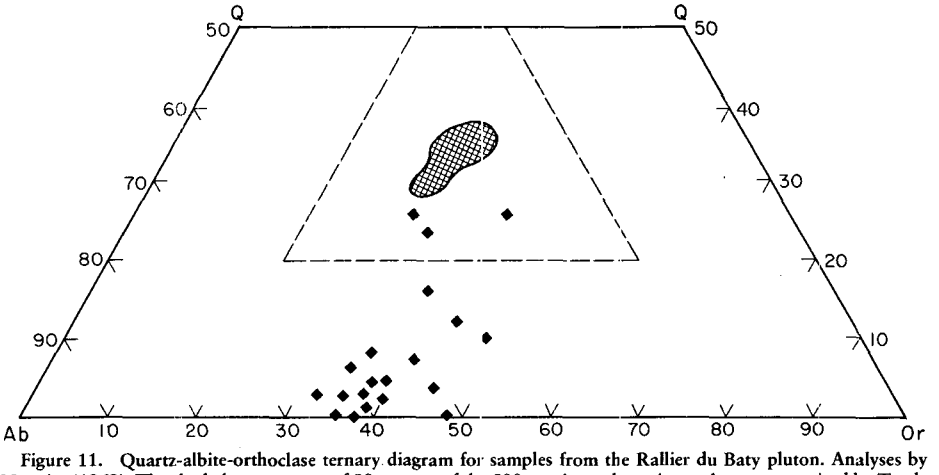


Figure 11. Quartz-albite-orthoclase ternary diagram for samples from the Rallier du Baty pluton. Analyses by Nougier (1969). The shaded area = zone of 90 percent of the 500 continental granite analyses summarized by Tuttle and Bowin (1958).

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values of 7 to 10 percent when the SiO_2 reaches 70 to 72 percent.

Thus the group appears to be a fractionated series, not unlike the commenditic trachyte and syenite found on several Atlantic islands (Schmincke, 1973).

Figure 11 shows the quartz-orthoclase-albite relations for the quartz-saturated parts of the pluton, and the limits of common continental granite (Tuttle and Bowin, 1958). There is no indication of strong affinity between these rocks and continental granite.

Isotope Geology

K-Ar Age Results. The results for the Foch Island tholeiite (Figs. 5 and 8) are presented in Table 3.

Argon extractions on −100 mesh powder did not indicate any diffusional loss of radiogenic argon, although the higher atmospheric contamination of these samples suggests that very fine grinding of specimens may be undesirable.

Table 3 shows that the three dated lava flows in the Foch Island section (Fig. 7) are indistinguishable within experimental error. The intrusive microsyenite stock at the base of the Grande Cascade section (Fig. 8) is slightly younger than the Foch Island basalt, but both dated sections are confined to the early Miocene.

We believe that our data, together with Molchanova's (1972) results, fix the age of the oldest parts of Kerguelen as early Miocene to late Oligocene. In view of the fact that the adjacent sea floor is older to the northeast and northwest, as discussed earlier, there is no simple evidence in the form of older rocks to support the microcontinental concept for Kerguelen.

Sr Isotopes. The initial Sr^{87}/Sr^{86} ratios of 0.7054 to 0.7073 (Table 4) are higher than those of many abyssal basalts (Peterman and Hedge, 1971). They are nevertheless within the range of results from some typical oceanic islands which show a progressive increase from 0.7020 to 0.7055 with increasing potassium content (Peterman and Hedge, 1971; Hedge and others, 1972). The potassium-rich Kerguelen syenite (Table 1) would therefore be expected, if it was oceanic, to have initial Sr^{87}/Sr^{86} ratios in the observed range 0.7050 to 0.7060. This is not uniquely diagnostic of oceanic origin, as igneous rocks erupted through segments of old continental crust usually have Sr^{87}-Sr^{86} values in the range 0.705 to 0.707.

The pluton is clearly not anomalously enriched in Sr^{87}. The results therefore do not resolve the problem of the Kerguelen microcontinental hypothesis: they are neither unambiguously consistent with remobilizing of sialic basement, nor do they exclude such a possibility. A study of the

TABLE 3. POTASSIUM-ARGON AGE DETERMINATIONS

Location	Body no.	K (%)	$Argon^{40*}$	Argon (atm)	Age (m.y.)	Mean age (m.y.)
Mount Bureau	4	0.517	5.56	26.8	26.7	
			5.57	26.4	26.8	26.5 ± 0.5
			5.38	38.8	25.9	
			5.56	31.5	26.8	
Mount Bureau	5	0.932	9.97	22.1	26.6	
			9.86	17.8	26.3	26.4 ± 0.5
			9.83	15.7	26.2	
Mount Bureau	7	0.348	3.72	58.4	26.6	
			3.88	55.1	27.7	27.3 ± 0.5
			3.85	52.9	27.5	
Le Grande Cascade	20	4.58	45.3	4.4	24.6	
			44.7	11.4	24.3	24.5 ± 0.5

See Figures 1, 5, 7, and 8 for geographic locations and stratigraphic details of sites (body numbers as in Figs. 7 and 8). $Argon^{40*}$ in CCNTP/gm x 10^7. Atmospheric argon in percent of total argon. Constants used are $\lambda\beta = 4.72 \times 10^{-10}$ yr^{-1}; $\lambda e = 5.84 \times 10^{-11}$ yr^{-1}; $^{40}K/K = 1.22 \times 10^{-4}$ g/g. See text for experimental methods.

TABLE 4. RUBIDIUM AND STRONTIUM ISOTOPE ANALYSES OF KERGUELEN PLUTONIC ROCKS

Sample no.	Rb (ppm)	Sr (ppm)	Rb/Sr	(Sr^{87}/Sr^{86})*	(Sr^{87}/Sr^{86})†	(Sr^{87}/Sr^{86})§
20T-5	7.1	338	0.021	0.7054	0.7054	0.7054
20E-5	110	180	0.61	0.7058	0.7056	0.7055
20F-8	182	131	1.39	0.7064	0.7058	0.7057
20F-15	208	56	3.71	0.7073	0.7059	0.7055

* *Observed Sr^{87}/Sr^{86}.*
† *Initial Sr^{87}/Sr^{86} at t = 9 m.y.*
§ *Initial Sr^{87}/Sr^{86} at t = 12 m.y.*

Location of samples: 20T-5 is a gabbro intrusive on the west-central Courbet Peninsula at Mount Vertes, immediately north of Val Studer, as indicated on Figure 2. 20E-5 and 20F-8 are shown on Figure 6; and 20F-15 is 400 m west of 20F-8.

TABLE 5A. PALEOMAGNETIC DATA (EACH UNIT)

Unit no.	N	D	I	R	K	α95	θ'	φ'
1	2	94.1	-61.8	1.921	12.6	77.65	28.8	193.3
2	3	4.8	-75.1	2.870	15.4	32.60	76.8	239.9
3	2	25.8	-51.8	1.996	269.5	15.27	64.5	128.5
4	3	23.0	-64.2	2.999	8,123.6	1.37	74.2	157.0
5	3	189.5	55.3	2.865	14.8	33.25	-75.0	280.0
6	3	135.2	62.9	2.952	41.6	19.36	-59.4	151.4
7	3	186.9	62.0	2.998	1,167.4	3.61	-82.4	291.2
8	4	354.7	-74.4	3.988	253.7	5.78	78.2	262.9
9	4	359.6	-78.0	3.992	365.9	4.81	72.3	250.8
10	4	329.8	-84.5	3.979	140.2	7.79	58.3	260.5
11	4	354.1	-78.4	3.981	157.4	7.35	71.3	257.2
12	5	23.5	-72.6	4.987	310.5	4.35	73.7	201.3
13*	3	50.7	-41.0	1.848	1.7		42.9	146.0
14	5	16.4	-30.4	4.993	596.7	3.14	54.5	97.9
15	4	9.7	-56.2	3.649	8.6	33.35	75.6	103.0
16	7	292.3	-54.7	6.991	644.8	2.38	39.7	329.6
17	6	9.2	-50.0	5.971	169.8	5.16	70.3	94.3
18	9	352.8	-62.0	8.981	415.9	2.53	82.2	28.1
19	6	73.6	-55.9	5.957	117.2	6.21	36.8	175.8
20	9	53.8	-67.9	8.885	69.6	6.21	56.2	184.0
21	3	339.6	-39.4	2.977	87.7	13.24	58.6	32.0
22	7	339.2	-64.4	6.896	57.6	8.02	75.7	344.5
23*	6	172.1	- 2.4	3.468	2.0	66.37	-39.0	240.0
24*	6	201.0	-17.5	3.587	2.1	66.39	-28.9	274.0
25	6	341.9	-50.0	5.955	110.1	6.41	67.0	27.1
26	6	321.7	-73.7	5.647	14.1	18.44	65.9	300.2
50	3	345.5	-41.2	2.981	105.6	12.06	61.9	41.0
51	2	354.7	-46.4	1.999	1,506.9	6.44	68.0	57.6

* *= 95% or greater probability of random directions.*

Explanation of symbols: Unit number refers to original field number of sampled lava flow or intrusive (see Figs. 4 and 5). N = number of separately oriented cores from each body. D and I = declination and inclination of remanent magnetism after demagnetization of each core in alternating magnetic fields of 100, 200, and 300 oersteds, and processing of results for minimum dispersion. D is in degrees east of north; I is in degrees with respect to horizontal (+ is downward; - is upward). R = resultant vector, applying unit vector per core; K = N-1/N-R, α95 = semivertical angle of 95% confidence cone; θ' and φ' = latitude and longitude of virtual geomagnetic pole.

TABLE 5B. STATISTICAL ANALYSIS OF PALEOMAGNETIC RESULTS (DIRECTIONS) BY SECTION

	B	N	D	I	R	K	θ'	φ'	dp	dm
(i)	28	128	5.6	-60.0	24.848	8.6	80.8	97.3	11.3	15.0
(ii)	22	98	359.7	-62.6	21.031	21.7	84.8	67.6	8.3	10.7

Explanation of symbols: B = number of separate bodies; N, D, I, R, K, θ', and φ' as in Table 5A, except unit vector per body (rather than unit vector per core) used throughout. dp and dm = angular lengths of major and minor semi-axes of 95% confidence oval around mean virtual geomagnetic pole, in line with and normal to great circle through sample site and virtual geomagnetic pole. (i) = all data. (ii) = excluding units for which θ' ≤45°.

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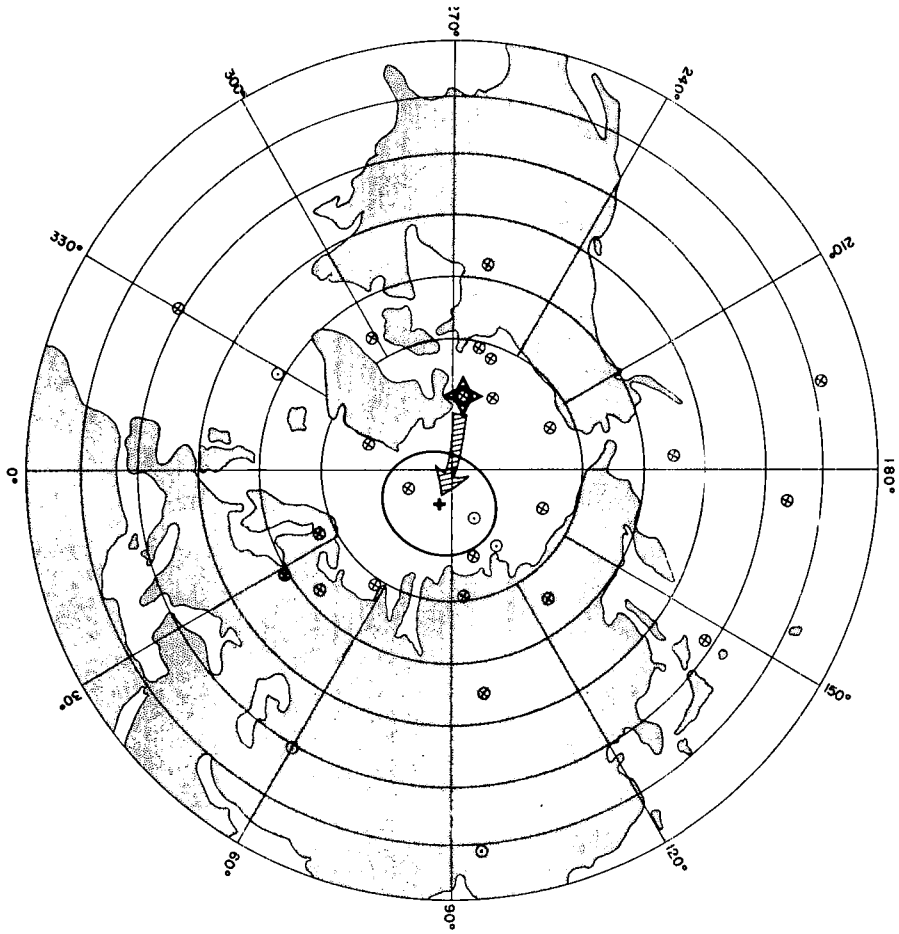


Figure 12. Virtual geomagnetic poles for each of the selected basalt flows (Figs. 5, 7, and 8). See Table 5a for details of data. ⊗ = Northern Hemisphere; ⊙ = reversed polarity, transposed to Northern Hemisphere. + = mean virtual geomagnetic pole, in center of 95 percent confidence oval (Table 5b). ✦ = position of 18.0 to 25 m.y. geomagnetic pole for Australia (McElhinny and Wellman, 1969; Wellman and others, 1969). Arrow is included to illustrate the preferred interpretation involving migration of the lower Miocene pole into that derived from the volcanic rocks of similar age on Kerguelen, by the process of crustal spreading of the island southward from the mid-oceanic ridge (Fig. 1).

Pb-isotopic composition may resolve the ambiguity.

The calculated initial Sr^{87}/Sr^{86} ratios for t = 9 and 12 m.y. (Table 4) allow a late Miocene age for the pluton, which is consistent with both stratigraphic relations and an unpublished commercial K-Ar date by Nougier (1969).

Paleomagnetism. When combined with the K-Ar results, the Foch Island paleomagnetic data show that a change in the geomagnetic polarity from the normal to the reversed sense occurred between t = 26 and 28 m.y. ago. None of the flows in this sequence has recorded a transitional polarity (which could be an indication of periods less than 10^3 to 10^4 yr between successive eruptions), but the K-Ar data suggest that immediately successive polarity epochs are very probably recorded.

Since the basal unit at the Grande Cascade section (Fig. 8) is also early Miocene (Table 3), it is clear that all of the paleomagnetic data can be combined to provide the first paleomagnetic pole determination for the period in the Indian Ocean region.

The mean virtual geomagnetic pole for all data combined is presented in Table 5 and Figure 12. Comparison of this result with a pole position for the same period from a reference region which is either stable or with known crustal movement can assist in analyzing crustal movement since the extrusion of the lavas involved. Fortunately, paleomagnetic pole positions for the same general period are available for two Australian sequences: the Nundewar Volcano (t = 17.5 ± 0.3 m.y.) and the southeast Queensland igneous rocks (t = 22 to 25 m.y.), as presented by Wellman and others (1969). The mean virtual geomagnetic poles for these two studies are added to Figure 12. Despite some scatter between flows, the mean Kerguelen paleomagnetic pole and associated confidence oval are completely consistent with migration southward away from the southeast branch of the mid-Indian ridge (Fig. 3) while the Australian rocks record a northward movement. A southward migration of Kerguelen and northward migration of Australia since t = 25 m.y. (to create the separation of mean VGP in Fig. 12) would correspond to a net crustal-spreading rate of about 4 to 6 cm yr^{-1}, which is in good agreement with both the crustal-spreading rates estimated from conventional marine magnetic anomaly analyses in the southeast Indian Ocean (Schlich and Patriat, 1971), and the inferred northward path of the Australian continent (Weissel and Hayes, 1972). While marine magnetic-anomaly patterns may be used to diagnose crustal spreading, they may not (for a restricted region) be capable of distinguishing between spreading from a stable mid-oceanic ridge system, and spreading involving migration of a ridge and sea floor away from a stable continental block. Combining the marine magnetic-anomaly data (Fig. 3) and the Kerguelen paleomagnetic result (Fig. 12) shows that the Kerguelen Plateau has migrated southward relative to Australia.

SUMMARY AND CONCLUSIONS

The island of Kerguelen, after a long history of building massive shield volcanoes of olivine and quartz tholeiite from late Oligocene to middle Miocene time, passed through a phase of alkali-basalt cone building associated with intrusion of trachyte dikes and syenite stocks, to eruptions of basanite flows and intrusion of phonolite plugs. A continuous petrochemical series is evident between quartz tholeiite, olivine tholeiite, alkali olivine basalt and basanite, and also between rhyolite, quartz trachyte, alkali trachyte, and phonolite. A "Daly Gap" separates the basic and felsic series, except on the Rallier du Baty Peninsula, where there is trachybasalt. It may be that rising magma was periodically trapped below such a thick lava pile. At a depth of perhaps 5 to 10 km, the residual liquid remaining after the magma was 95 percent crystallized may remain liquid for tens of thousands of years. Earth movements may finally allow the lower density residual liquid to be emplaced near the surface, or extruded. The residual liquid from quartz tholeiite was rhyolite, that of olivine tholeiite was quartz trachyte, that of alkali basalt was trachyte, and of basanite was phonolite.

Volcanic activity appears to have migrated from north to south with time, but the concentration of more silicic lava to the northwest and more alkaline to the southeast, though suggestive of different depths of magma generation, cannot be explained in terms of what is presently known of the tectonic patterns in the region. The general structure in Kerguelen of massive tholeiitic shield volcanoes with later

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alkaline lava and basanite-phonolite culminations is similar in some respects to the Hawaii sequence (Macdonald and Katsura, 1964), although Hawaii lacks rhyolite. Tholeiitic basalt and rhyolite flows grading to alkali basalt and hawaiite are found in Iceland (Jakobsson, 1972), but highly alkaline members have not been reported; on Ascension Island, quartz trachytes and commendites with silica content increasing toward rhyolite composition have been reported (Mitchell-Thomé, 1970), as on Kerguelen.

The Heard Island volcanic rocks, on the southern end of the Kerguelen-Heard plateau, consist mainly of ankaramite, alkali basalt, trachybasalt, trachyte, and phonolite (Stephenson, 1964). The association is similar to the younger alkali basalt, trachybasalt-trachyte-phonolite suite of Kerguelen. Stephenson (1964) found some of the Heard basalt flows to be hypersthene normative. It is possible that the submerged base of Heard Island is a tholeiite shield as seen in Kerguelen.

The early Miocene paleomagnetic data for Kerguelen are compatible with a southward migration of the Kerguelen Ridge away from the mid-oceanic ridge system. The Kerguelen-Gaussberg Ridge has not been a stable block with respect to the adjacent mid-oceanic ridge system.

The maximum isotopic age so far defined for any outcrop in Kerguelen is early Miocene to late Oligocene. This is at least 15 to 20 m.y. younger than the sea floor to the north and northeast of the ridge, as defined by marine magnetic-anomaly analysis. The Rb- and Sr-isotopic data for the Rallier du Baty pluton show that it is probably late Miocene in age, and while neither confirming nor denying the possibility of the body being a result of remobilized continental crust, together with the occurrence of syenites (the genesis of which has never been associated with remobilized continental crust), strongly suggest an oceanic origin for the pluton.

Our preferred interpretation of the petrochemistry, volcanic stratigraphy, and the paleomagnetic and isotopic data for Kerguelen is that it is a simple oceanic island located over a region of large-scale magma generation. There is no evidence suggesting the presence of continental material on the island. This preferred interpretation nevertheless cannot entirely eliminate the possibility of a microcontinental fragment beneath the Kerguelen Ridge. If it does exist, the effect on the island surface features has been minimal. It may be necessary to reconsider the origin of Kerguelen Ridge and Broken Ridge. Even if they existed prior to the initiation of the latest phase of crustal spreading in the southeast Indian Ocean, there is no reason why these very large ridges cannot be essentially oceanic; they could, for example, be associated with the earliest phases of oceanic crustal generation, prior to the beginning of regular large-scale production of the sea floor. If this was the case, then the marine sediments of the Kerguelen-Gaussberg Ridge would yield early Tertiary or even Mesozoic microfauna.

Sampling by drilling may provide samples capable of resolving any ambiguity which still remains concerning the origin of the Kerguelen-Gaussberg Ridge. Our study suggests that the cores recovered could be as old as Mesozoic, but "basement" rocks will not include true continental materials.

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